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$$\begin{aligned}2 \log_4 x + \frac{1}{\log_{x-4} 2} &= 2 + \frac{1}{\log_{x-3} \sqrt{2}} \\ \Leftrightarrow 2 \frac{\log_2 x}{\log_2 4} + \log_2 (x-4) &= 2 + \frac{1}{2} \log_{\sqrt{2}} x - 3 \\ \Leftrightarrow 2 \frac{\log_2 x}{\log_2 4} + \log_2 (x-4) &= 2 + \frac{\log_2 (x-3)}{2 \log_2 \sqrt{2}} \\ \Leftrightarrow \log_2 x + \log_2 (x-4) &= \log_2 4 + \log_2 (x-3) \\ \Leftrightarrow \log_2 (x(x-4)) &= \log_2 (4(x-3)) \\ \Leftrightarrow x(x-4) &= 4(x-3) \\ \Leftrightarrow x^2 - 4x &= 4x - 12 \\ \Leftrightarrow x^2 - 8x + 12 &= 0\end{aligned}$$

D vinden

$$D = 64 - 4 \cdot 12$$

$$D = 16$$

$$x_1 = \frac{8+4}{2} = 6$$

$$x_2 = \frac{8-4}{2} = 2$$

Merk op $\log_{x-4}$ dus $x > 4 \Rightarrow x = 6$

References